

Large- n Limit of $SU(n)$ Quantum "Spin" Chains

Ian Affleck^(a)

Service de Physique Théorique, Centre d'Etudes Nucléaires de Saclay, F-91191 Gif-sur-Yvette Cedex, France

(Received 13 December 1984)

A class of $SU(n)$ antiferromagnetic quantum "spin" chains is solved in the large- n limit. While the low-lying spectrum of the normal spin chain ($n=2$) consists of massless solitons, in the large- n limit the soliton is massive. Furthermore, the models exhibit many features of the CP^n field theories at nonzero topological charge, which, it was recently argued, they become in the limit of a large representation.

PACS numbers: 11.10.Lm, 05.50.+q, 75.10.Jm

While the Bethe *Ansatz* solution of the antiferromagnetic quantum spin- $\frac{1}{2}$ chain has existed for many years,¹ it was only recently observed that the low-lying spectrum consists of massless spin- $\frac{1}{2}$ solitons (with no soliton bound states).² In another interesting development Haldane³ argued that the large-spin limit of the model with minimal interaction,

$$H = \sum_r \mathbf{S}_r \cdot \mathbf{S}_{r+1},$$

is the $O(3)$ σ model with topological angle (θ) zero or π for integer or half-integer spin. By addition of a staggered interaction θ can be varied continuously. If the correspondence remains qualitatively correct even for finite spin, one expects the integer-spin chains to exhibit a mass gap (as does the σ model at $\theta=0$) and the higher half-integer-spin chains and the σ model at $\theta=\pi$ to be massless. This correspondence was extended in an earlier work⁴ where generalized "spin" chains were introduced with generators of a representation (rep) of $SU(n)$ on even lattice sites and generators of the conjugate rep on odd sites. It was argued that the rep with a rectangular Young tableau of m rows and N columns gives the $U(n)/U(m) \otimes U(n-m)$ σ model in the limit $N \rightarrow \infty$. Again θ is zero or π for N even or odd and can be varied continuously by addition of a staggered interaction. These σ models are known⁵ to have a mass gap (for all θ) in the limit $n \rightarrow \infty$, m fixed. But since the $m=1$, $n=2$ model is the $O(3)$ model, it seems likely that the mass gap vanishes at $\theta=\pi$ for some range n, m . In the limit $n, m \rightarrow 0$ this critical point may describe the localization transition in the quantum Hall effect.⁶ Unfortunately, this picture seems questionable in light of an exactly integrable $SU(2)$ spin- s chain (with an interaction which is a $2s$ -order polynomial in $\mathbf{S}_r \cdot \mathbf{S}_{r+1}$) which has no mass gap for half-integer or integer spin.⁷

In this Letter we solve the $SU(n)$ chain, for a Young tableau of one row with length N , in the limit N fixed, $n \rightarrow \infty$. As for original spin- $\frac{1}{2}$ model (which our model becomes at $N=1$, $n=2$), the low-lying

spectrum consists of solitons (and antisolitons) in the n (and $\bar{n}$) of $SU(n)$. Unlike the original model, the solitons have a transparent physical interpretation (they interpolate between two vacua resulting from the spontaneous breakdown of parity) and a nonzero mass. Thus the solitons must develop a mass at some critical value of n . Furthermore there is a weak [$O(1/N^2)$] linear confining potential between solitons and antisolitons for N even, but the solitons are liberated for N odd. The system also moves periodically through a deconfining phase as a function of the alternating coupling. This behavior is analogous to that of the $CP^{(n)}$ model, which at large n becomes massive QED with a weak gauge coupling and external charges $\pm\theta/2\pi$. The linear confining potential between electrons and positrons vanishes at $\theta=(2p+1)\pi$ (for integer p). Thus this work supports the "spin"-chain σ model identification and suggests that the exactly integrable spin Hamiltonians may have a different low-energy spectrum (at least for integer spin) than the minimal Hamiltonian.

The model⁴ has matrices S_j^i ($i, j=1, \dots, n$) generating the symmetric N -tensor rep of $SU(n)$ on even sites, and matrices $\bar{S}_j^i$ generating the conjugate rep on odd sites. These matrices obey the $SU(n)$ commutation relations

$$[S_j^i, S_k^j] = \delta_j^i S_k^j - \delta_j^k S_i^j,$$

and the condition $S_i^i = 0$ (repeated indices are summed). The Hamiltonian is

$$H = -\frac{1}{n} \sum_{r, \pm} (S_r^j)_{2r} (\bar{S}_r^j)_{2r \pm 1}.$$

It is very convenient to use a fermionic representation for the "spin" variables:

$$S_j^i = \psi^{\dagger i} \psi_j - \delta_j^i / n.$$

The normal fermion anticommutation relations $\{\psi^{\dagger i}, \psi_j\} = \delta_j^i$ imply the correct "spin"² commutation relations. However, the Hilbert space of the fermion is too large. At each site it consists of the vacuum (singlet), one-particle states (n), two-particle states



FIG. 1. A typical singlet state.



FIG. 2. Ground state with two sites.

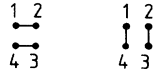


FIG. 3. The two singlet states with four sites.

(antisymmetric two-tensor rep), $\dots$, $(n-1)$ -particle states ($\bar{n}$), and the n -particle state (singlet). We must project out the one-particle state to reproduce the spin system correctly, i.e., we impose

$$\psi^\dagger_i \psi_i |\phi\rangle = |\phi\rangle.$$

On odd sites we project out the $(n-1)$ -particle states and write

$$\bar{S}_j^\dagger = \psi^\dagger_j \psi_j - \delta_j^i (n-1)/n.$$

On odd sites it is more convenient to think of ψ_i as a hole creation operator:

$$\bar{\psi}_i^\dagger \equiv \psi_i,$$

and set

$$\bar{S}_j^\dagger = -[\bar{\psi}_j^\dagger \bar{\psi}_j - \delta_j^i / n]$$

with a projection onto one-hole states

$$\bar{\psi}_i^\dagger \bar{\psi}_i |\phi\rangle = |\phi\rangle.$$

The Hamiltonian can now be written

$$H = -\frac{1}{n} \sum_{r,\pm} \psi_{2r}^\dagger \psi_{j,2r} \bar{\psi}_{i,2r\pm 1}^\dagger \bar{\psi}_{2r\pm 1}^j.$$

To obtain the symmetric N -tensor rep we introduce N "flavors" as well as n "colors" of fermions ψ_{if} ($f=1, \dots, N$) and project out the flavor-singlet N -particle or N -hole subspace

$$S_j^\dagger = \psi^\dagger_{jf} \psi_{jf} - \delta_j^i N/n, \quad \psi^\dagger_{if} \psi_{ig} |\phi\rangle = \delta_g^f |\phi\rangle,$$

$$\bar{S}_j^\dagger = -[\bar{\psi}_{jf}^\dagger \bar{\psi}_{jf} - \delta_j^i N/n], \quad \bar{\psi}_{if}^\dagger \bar{\psi}_{ig} |\phi\rangle = \delta_g^f |\phi\rangle.$$

We now consider the fundamental rep ($N=1$). A state of the system is specified by the color of the electron or hole at each site. The term in H which connects two neighboring sites can only act if the electron and hole have the same color (j). In that case it maps the state into a sum over i of the state with the same color, i . That is,

$$H|jk\rangle = -(\delta_k^j/n) \sum_i |ii\rangle.$$

Color-singlet states have pairwise contractions of all particle and hole indices. We represent such states diagrammatically by bonds connecting even to odd sites whose indices are contracted (Fig. 1).

To warm up, let us consider the problem with only two sites. All states with different colors on the two sites have energy zero. Of the n remaining states,

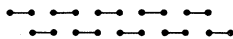


FIG. 4. The two ground states.

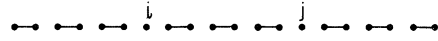


FIG. 5. A soliton-antisoliton state.

there are $n-1$ of the form $|ii\rangle - |11\rangle$ ($i=2, \dots, n$) also with energy zero, and finally the singlet state $n^{-1} \sum_i |ii\rangle$ with energy -1 . The singlet is the ground state (Fig. 2). Now consider a closed chain with four sites. The lowest-energy states are again singlets but now there are two of them (Fig. 3) corresponding to the two ways of connecting even to odd sites. Denoting them by the paired sites $|12, 34\rangle$ and $|14, 32\rangle$ and denoting the four terms in H by the pairs of sites on which they act, we see that

$$H_{12}|12, 34\rangle = H_{34}|12, 34\rangle = -|12, 34\rangle$$

but

$$H_{23}|12, 34\rangle = H_{41}|12, 34\rangle = -n^{-1}|14, 32\rangle.$$

Thus the ground state is the symmetric combination $\frac{1}{\sqrt{2}}[|12, 34\rangle + |14, 32\rangle]$. However, in the large- n limit the two states $|12, 34\rangle$ and $|14, 32\rangle$ are almost degenerate; the mixing is $O(1/n)$.

It is now easy to take the thermodynamic limit at large n . To leading order in $1/n$, H simply counts the number of nearest-neighbor bonds. There are two degenerate ground states $|e0\rangle$ and $|0e\rangle$ with bonds between sites $2r$ and $2r+1$ or $2r-1$ and $2r$, for all r (Fig. 4). $|e0\rangle$ is an eigenstate of $H_{2r, 2r+1}$ (for all r). It is *not* an eigenstate $H_{2r-1, 2r}$; however,

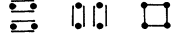
$$H_{2r-1, 2r}|e0\rangle = O(1/n).$$

For a system with a finite number of sites, N_s , these states are mixed by an amount of order $(1/n)^{N_s}$; in the thermodynamic limit there are two ground states. Parity (reflection about a site) is spontaneously broken.

It is also easy to find the excitation spectrum. Decreasing the number of nearest-neighbor bonds by one gives states with an uncontracted lower index at some site and an uncontracted upper index at some other site (Fig. 5). This clearly corresponds to a soliton-antisoliton pair. The soliton and antisoliton are in the n and $\bar{n}$; the pair, in the singlet or adjoint rep. (If N_s is even solitons and antisolitons must occur in pairs. If N_s is odd the "ground state" contains a soliton.) Since we lost only one nearest-neighbor bond to produce a soliton-antisoliton pair, the energy of a soliton is $\frac{1}{2}$. (This is very reminiscent of "charge fractionalization" in polyacetylene.⁸) To lowest order in $1/n$ a static soliton $|r\rangle$ is an eigenstate; to next order we must consider a momentum eigenstate

$$|k\rangle = \sum_r e^{ikr} |r\rangle.$$

FIG. 6. Ground state for $N=2$ and two sites.

FIG. 7. Singlet states for $N=2$ and four sites.

Since $H_{r,r\pm 1}|r\rangle = -n^{-1}|r\pm 1\rangle$, we find

$$E(k) \approx \frac{1}{2} - (2/n) \cos k\Delta$$

$$\xrightarrow{k \rightarrow 0} \left(\frac{1}{2} - 2/n\right) + k^2 \Delta^2/n. \quad (1)$$

(Δ is the lattice spacing.) The solitons are nonrelativistic with a mass of $O(n)$.

Let us now consider the symmetric two-tensor rep ($N=2$). There are now two symmetric color indices at each site. All singlet states now have two bonds ending at each site. For the two-site system the ground state is again the unique singlet (Fig. 6). With four sites there are three singlet states (Fig. 7), two with double bonds and one with single bonds. Again the true ground state is a linear combination but the mixing is $O(1/n)$. Passing to the thermodynamic limit there are now, to leading order in $1/n$, three ground states, two with entirely double bonds (and broken parity) and one with only single bonds (Fig. 8). Of course, since no symmetry connects the third state to the first two, we do not expect this degeneracy to be exact.

To determine the true ground state we use Rayleigh-Schrödinger perturbation theory in $1/n$. As unperturbed Hamiltonian we take the operator that counts the number of nearest-neighbor bonds.

In second-order perturbation theory

$$\delta E = \sum_{i \neq 0} \frac{|\langle i|H'|0\rangle|^2}{E_i - E_0}. \quad (2)$$

We must consider the mixing of the ground states with a soliton-antisoliton pair (Fig. 9). For the symmetric ground state we have

$$\langle s\bar{s}|H_{r,r+1}|0\rangle = -2/n \quad (r \text{ even or odd}),$$

whereas for the $|e0\rangle$ parity-breaking state, we have

$$\langle s\bar{s}|H_{2r,2r-1}|0\rangle = -2/n, \quad \langle s\bar{s}|H_{2r,2r+1}|0\rangle = 0.$$

Thus,

$$\delta E = -(2/n)^2 N_s \quad \text{for the symmetric state,}$$

$$= -(2/n)^2 N_s/2 \quad \text{for the parity-breaking states.}$$

The real ground state preserves parity.

The excitation spectrum again consists of soliton-

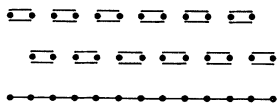
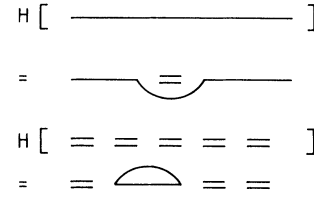
FIG. 8. Ground states for $N=2$.

FIG. 9. Mixing of ground state with soliton-antisoliton state.

antisoliton pairs which interpolate between the real and false vacua. However, there is now a small linear potential between soliton and antisoliton because of the difference in energy of the vacua:

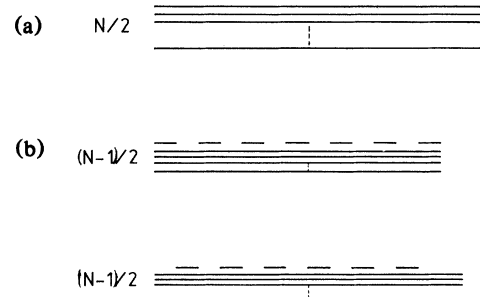
$$E = (2/n^2)L,$$

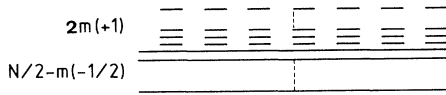
where L is the length of the region in the false vacuum. The true spectrum is given by quantizing the $\bar{s}s$ separation. At low energies one obtains an effective continuum Hamiltonian for a particle in a linear potential.

It is now clear what happens for any fixed value of N at $n \rightarrow \infty$. If N is odd the ground state breaks parity and the spectrum consists of free solitons (Fig. 10). If N is even parity is unbroken and the soliton-antisoliton pairs feel a linear confining potential $\propto 1/n^2$.

This is the qualitative behavior of the $CP^{(n)}$ model at large n .⁵ The model is then equivalent to QED with a weak coupling $e^2 \propto 1/n$. The topological angle, θ , corresponds to external charges $\pm e\theta/2\pi$ at $\pm\infty$. At $\theta=0$ there is a linear confining potential ($\sim e^2$) between electrons and positrons. However, at $\theta=\pi$ this force disappears provided that electrons and positrons alternate so as to flip the sign of the net electric field.⁹ The agreement is only qualitative, as should be expected for finite N . The soliton kinetic energy of Eq. (1) does not agree with the relativistic $CP^{(n)}$ model with $M_s \propto \exp(-2\pi n/g^2) \sim \exp(-8\pi n/N)$.⁵

Finally, let us examine the effect of a staggered in-

FIG. 10. (a) Ground state for N even; (b) ground state for N odd.

FIG. 11. Quasidegenerate ground states for N even (odd).

teraction

$$H = -\frac{1}{n} \sum_r \left[\frac{1 \pm \alpha}{n^2} \right] S_{2r} \bar{S}_{2r \pm 1}.$$

The additional interaction changes the energy of an $e0$ or $0e$ bond to $-(1 \pm \alpha/n^2)$. Thus it favors unsymmetric states (it explicitly breaks parity). However, it must compete with second-order perturbation theory for the original Hamiltonian [we have chosen it to be of the same order, $O(1/n^2)$] which prefers symmetric states. For arbitrary N the quasidegenerate ground states have $N/2 - m$ lines and $2m$ dashes (N even) or $N/2 - \frac{1}{2} - m$ lines and $2m + 1$ dashes (N odd) (see Fig. 11). The energy of these states in second-order perturbation theory [Eq. (2)] is determined by the amplitude for H to produce a soliton-antisoliton pair. When H acts on a link with p bonds in between two links with q bonds, this amplitude is

$$H|0\rangle = -n^{-1}q^2(p+1)^{1/2}q^{-1}|\bar{s}\bar{s}\rangle.$$

The factor of q^2 arises from the choice of which of the q bonds to annihilate on the neighboring links; the other factors arise from the $1/\sqrt{p!}$ normalization in the wave function for a link with p bonds. Specializing to the case $m \ll N \ll n$ we find

$$\begin{aligned} \frac{E(m)}{N_s} &= \text{const} + \frac{1}{n^2} \left[\frac{m^2 N}{2} - \alpha m \right] \quad (N \text{ even}), \\ &= \text{const} + \frac{1}{n^2} \left[\frac{(m + \frac{1}{2})^2 N}{2} - \alpha m \right] \quad (N \text{ odd}). \end{aligned}$$

We see that the true ground state changes (one bond moves from each $e0$ site to the neighboring $0e$ site) each time

$$\alpha = (m + \frac{1}{2})N \quad (N \text{ even}),$$

or

$$\alpha = mN \quad (N \text{ odd}).$$

At these values of α two ground states are degenerate so that the solitons are liberated. The same phenomena occurs in the $CP^{(n)}$ model. Each time $\theta = (m + \frac{1}{2})2\pi$ the ground state changes (a pair of charges are produced and go to $\pm\infty$ to screen the

external charges) and the electrons and positrons are liberated (provided they alternate as must solitons and antisolitons). Here, effectively

$$\theta = \pi N(1 + \alpha/N^2).$$

This linear dependence of θ on α was also predicted in Ref. 4. It was argued in Ref. 4 that this first-order transition in the large- n $CP^{(n)}$ model may become second order at $n=1$ [and more generally for some range of n and m near zero in the $U(n)/U(m) \otimes U(n-m)$ models]. The fact that the mass gap for the fundamental-rep spin chain vanishes for some range of n near zero gives qualitative support to this picture.

In conclusion, the large- n limit considered here gives a simplified picture of the original spin- $\frac{1}{2}$ chain. It also exhibits much similarity with the large- n $CP^{(n)}$ model and thus supports the arguments of Refs. 3 and 4.

I would like to thank E. Brézin for several useful discussions. I acknowledge receipt of a fellowship from the A.P. Sloan Foundation.

(a)On leave from Physics Department, Princeton University, Princeton, N.J. 08544.

¹H. Bethe, Z. Phys. **71**, 205 (1931).

²N. Andrei and J. Lowenstein, Phys. Rev. Lett. **43**, 1698 (1979); L. Faddeev and L. Takhtajan, Phys. Lett. **85A**, 375 (1981).

³F. Haldane, Phys. Lett. **93A**, 464 (1983), and Phys. Rev. Lett. **50**, 1153 (1983), and to be published.

⁴I. Affleck, CEN, Saclay, Report No. SPhT84/144, 1984 (to be published).

⁵A. D'adda, P. di Vecchia, and M. Luscher, Nucl. Phys. **B146**, 63 (1978); E. Witten, Nucl. Phys. **B149**, 285 (1979); E. Brézin, S. Hikami, and J. Zinn-Justin, Nucl. Phys. **B165**, 528 (1980).

⁶H. Levine, S. Libby, and A. Pruisken, Phys. Rev. Lett. **51**, 1915 (1983); Nucl. Phys. **B240**, 30, 4971 (1984); A. Pruisken, Nucl. Phys. **B235**, 277 (1984), and to be published.

⁷P. Kulish and E. Sklyanin, in *Integrable Quantum Field Theories*, edited by H. Araki *et al.*, Lecture Notes in Physics, Vol. 151 (Springer, New York, 1982), p. 61; P. Kulish, N. Yu. Reshetikhin, and E. Sklyanin, Lett. Math. Phys. (Netherlands) **5**, 393 (1981); L. Takhtajan, Phys. Lett. **87A**, 479 (1982); G. Babudjian, Phys. Lett. **90A**, 479 (1982), and Nucl. Phys. **B215**, 317 (1983).

⁸W. P. Su, J. R. Schrieffer, and A. J. Heeger, Phys. Rev. Lett. **42**, 1698 (1979), and Phys. Rev. B **22**, 2099 (1980).

⁹S. Coleman, Ann. Phys. (N.Y.) **101**, 239 (1976).